



Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • goal: uncover spending patterns, segments, preferences, subscriptions

Dataset Summary

Size

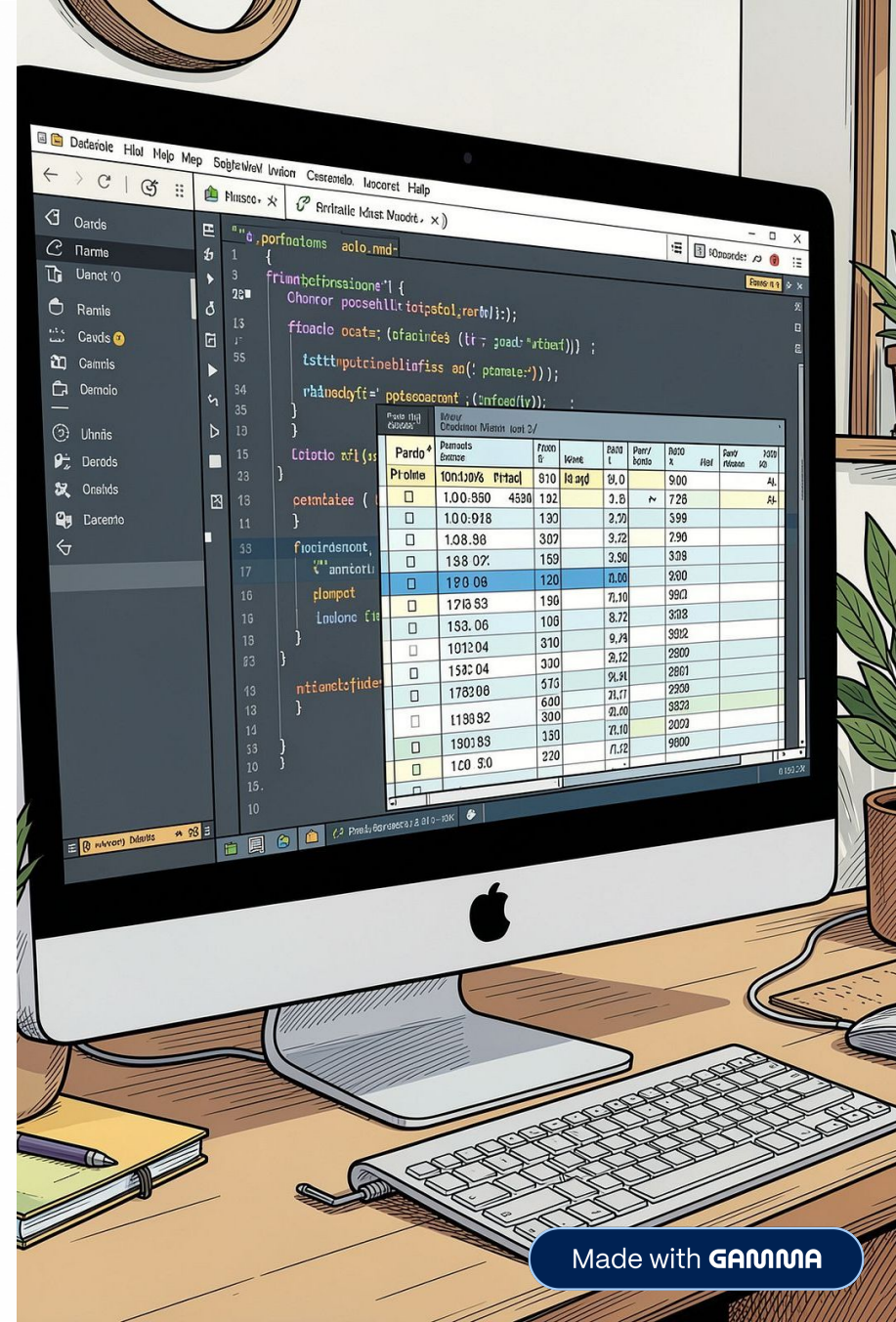
Rows: 3,900 • Columns: 18

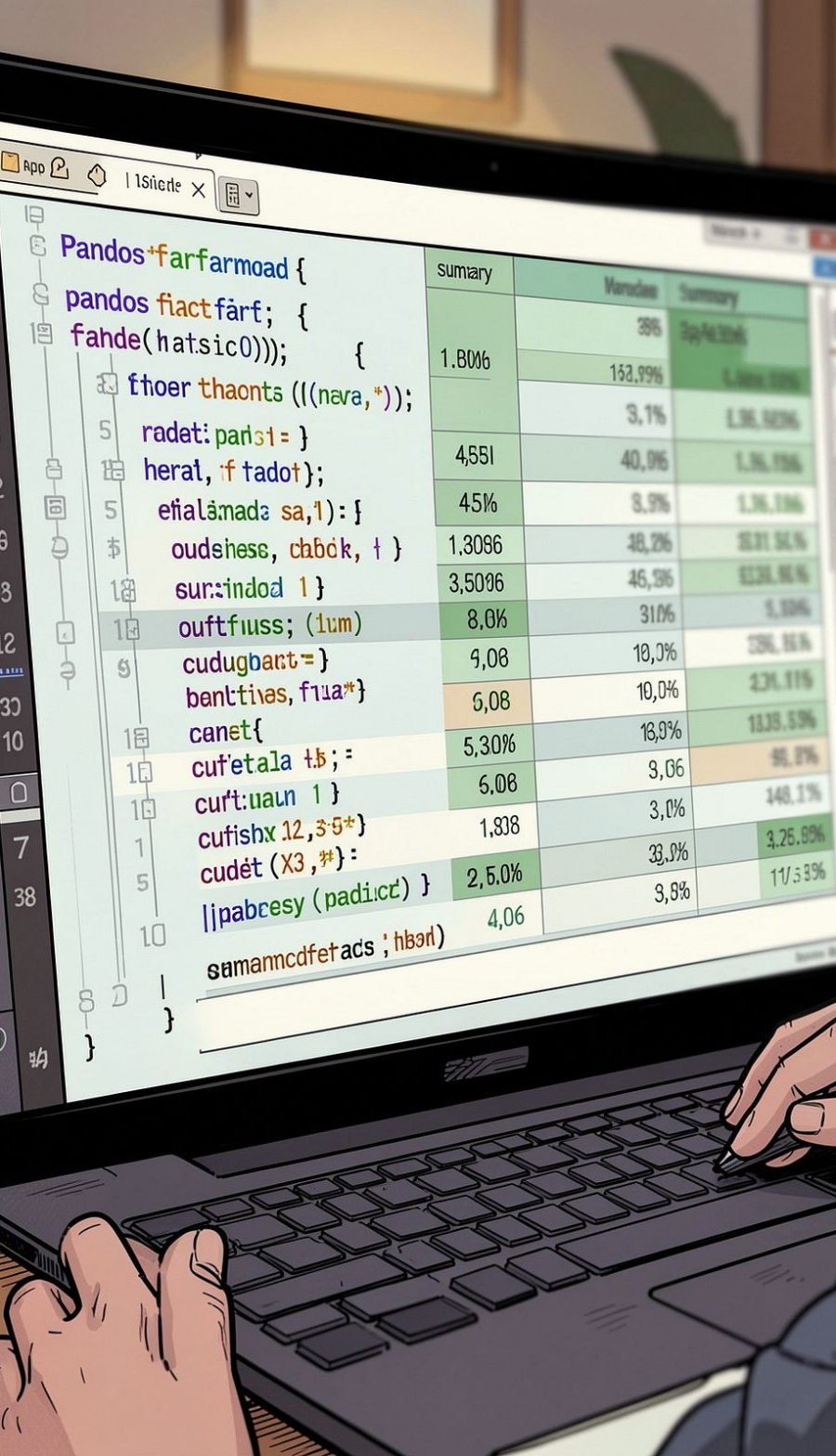
Key Features

Demographics, Purchase details,
Behavior, Shipping, Discounts

Missing Data

Review Rating: 37 nulls





Exploratory Data Analysis (Python)

01

Load & Inspect

`df.info()` • `df.describe()`

02

Clean

Imputed review_rating by category median • renamed to snake_case

03

Feature Eng.

`age_group` • `purchase_frequency_days`

04

DB Integration

Loaded cleaned DF into PostgreSQL



Revenue (yqq)	Semnue	GenGer	Genger
\$1 \$b/6oncank / % → \$0L	57,93	\$42	\$,73
1.1 flay/Narientb /			
\$1 avey 6noi / Lt % → \$0L	19,23	3.63	\$5,45
\$4 \$k/fooe:/ / % → \$0L	49,80	0.78	903
\$6 flu/nogty:/ / % → \$0L	00:10	47.38	632
\$0 \$h/6ongc:/ / % → \$0L	41.68	2.45	5P67
\$3 flu/oog:/ / % → \$0L	11,72	1.31	38,33
\$0 \$b/6engr:/ / % → 09.81)	49,63	46,5	44,06
\$1 \$h/acLenk / % → 09.0/10	49,66	36,11	45,72
1.1 flu6 bluaqan /			
\$3 #tp/lag/Lt / / % → \$8/73	39,43	48.4	19,68
\$6 \$tp/coni:/ / % → J06266)	39,88	43.6	36,89
\$0 flu/soht:/ / % → \$4/09	45,01	751	38,06

Revenue by Gender

Male: 157,890 • Female: 75,191



Male customers generated higher total revenue



Discounts & High-Spenders

839 rows: customers who used discounts yet spent above average

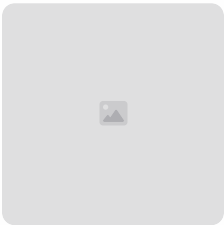
Discount-Dependent Items

Top: Hat, Sneakers, Coat, Sweater, Pants

Policy Check

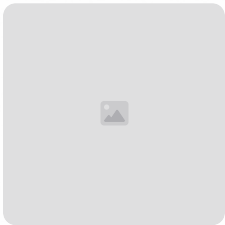
Review discount impact on margins

Top Products by Rating



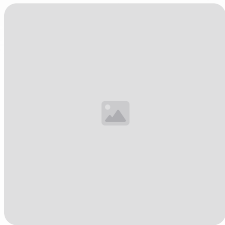
Gloves

Avg Rating: 3.86



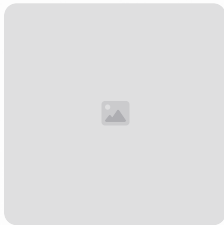
Sandals

3.84



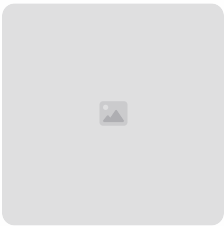
Boots

3.82



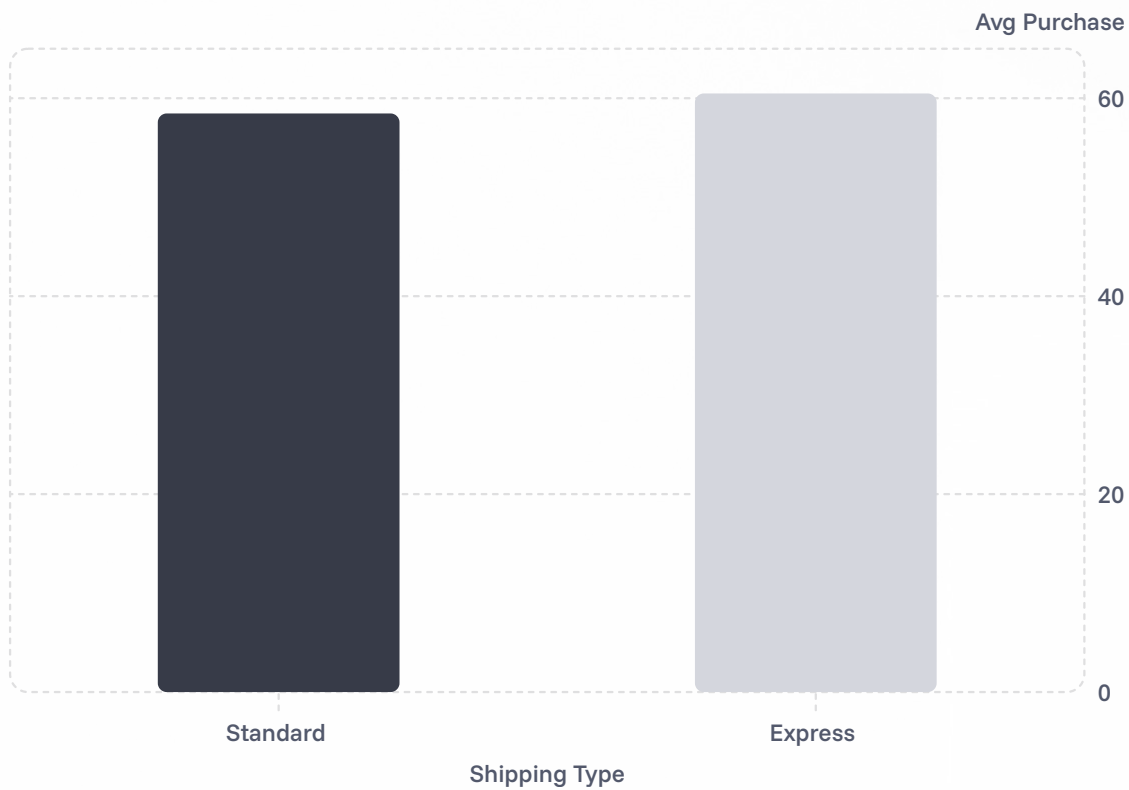
Hat

3.80



Skirt

3.78



Shipping & Spend

Express slightly higher avg purchase (\$60.48 vs \$58.46)

📌 Consider targeting express-shipping users



Customer Segments & Loyalty



Loyal

3,116 customers



New

83 customers



Returning

701 customers

Repeat buyers (>5 purchases): more likely to subscribe

Power BI Dashboard Highlights

Key metrics: Customers 3.9K • Avg Purchase \$59.76 • Avg Rating 3.75

- **Revenue by Category**

Clothing 100K •
Accessories 75K •
Footwear 30K • Outerwear
15K

- **Sales by Category**

Clothing 1,800 •
Accessories 1,200 •
Footwear 600 • Outerwear
200

- **Age Group Revenue**

Young Adult 62,143 • Middle-aged 59,197 • Adult 55,978 •
Senior 55,763



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow 'Loyal' segment

Discount Policy

Balance sales lifts with margin control

Product Positioning

Highlight top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

Next steps: prioritize subscription growth & margin-focused promotions